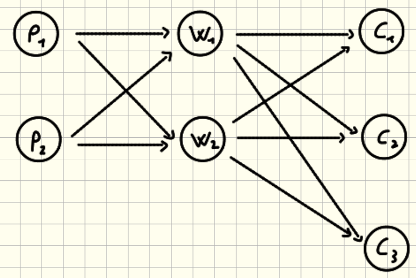
Problema di trasporto



Costi (K_1)	W_1	W_2
P_1	1	6
P_2	4	2

Costi (K_2)	C_1	C_2	C_3
W_1	3	4	5
W_2	2	1	2

Domande $\underset{\times 10^3}{d} = (50,\ 100,\ 50)$ Capacità $\underset{\times 10^3}{f} = (150, 60)$

$P = \{P_1, P_2\}$ $W = \{W_1, W_2, W_3\}$ $C = \{C_1, C_2, C_3\}$

$X(p,w) \geq 0,\ Y(w,c) \geq 0$ con $p \in P,\ w \in W,\ c \in C$

$$\min Z = \sum_{p \in P} \sum_{w \in W} K_1(p,w)\, X(p,w) + \sum_{w \in W} \sum_{c \in C} K_2(w,c)\, Y(w,c)$$

vincoli: $\sum_{w \in W} X(p_i, w) \leq F_i \quad \forall p_i \in P \longrightarrow \left.\begin{array}{l} X(p_1,w_1) + X(p_1,w_2) \leq 150\,000 \\ X(p_2,w_1) + X(p_2,w_2) \leq 60000 \end{array}\right\} \Rightarrow$ <u>Vincoli di capacità</u>

$\sum_{c \in C} Y(w_i, c) = d_i \quad \forall w_i \in W \longrightarrow \left.\begin{array}{l} Y(w_1,c_1) + Y(w_2,c_2) = 50\,000 \\ Y(w_1,c_2) + Y(w_2,c_2) = 100\,000 \\ Y(w_1,c_3) + Y(w_2,c_3) = 50\,000 \end{array}\right\} \Rightarrow$ <u>Vincoli di domanda</u>

$\sum_{p \in P} X(p,w) = \sum_{c \in C} Y(w,c) \longrightarrow \begin{array}{l} X(p_1,W_1) + X(p_2,W_1) = Y(W_1,C_1) + Y(W_1,C_2) + Y(W_1,C_3) \\ X(p_1,W_2) + X(p_2,W_2) = Y(W_2,C_1) + Y(W_2,C_2) + Y(W_2,C_3) \end{array}$

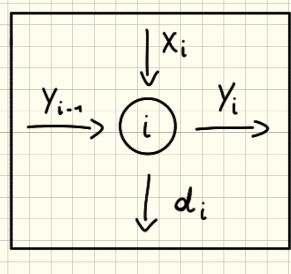
$\forall w \in W$ <u>Vincoli di flusso</u>

Pianificazione multiperiodo $t = \{1, 2, 3, 4, 5\}$

$d_t = \{30, 60, 40, 70, 60\}$ $C_t = \{8, 8, 10, 10, 20\}$ $h_t = \{1, 1, 2, 2, 2\}$ Mag. iniziale: 0

X_t = quantità prodotta al tempo t Y_t = quantità in magazzino al tempo t

$$\min Z = \sum_{t=1}^{5} C_t X_t + \sum_{t=1}^{5} h_t Y_t \qquad Y_0 = 0$$



vincoli: $X_i + Y_{i-1} - d_i - Y_i = 0$

$X_i \geq 0,\ Y_i \geq 0 \qquad \forall i \in t$